

Week 2: Games with Sequential Moves

Backward Induction, Credibility, and Timing Advantage

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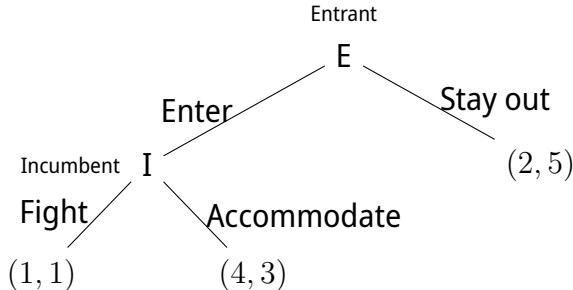
Semester 2026–27 Term 1

When later players can observe and respond, timing changes the game.

Opening Decision: Enter or Stay Out?

The Situation

A new firm decides whether to **Enter** a market. If it enters, the incumbent observes entry and then chooses whether to **Fight** or **Accommodate**.



What should the entrant do?

The entrant's payoff is listed first; the incumbent's payoff is second.

Today's Learning Outcomes

By the end of class, you should be able to:

- 1 distinguish simultaneous from sequential strategic interaction;
- 2 read and construct a simple **game tree**;
- 3 distinguish an **action** from a complete strategy;
- 4 solve a sequential game using **backward induction**;
- 5 evaluate credibility and first- or second-mover advantage from the structure of the game.

From Week 1 to Week 2

Week 1: Strategic Interaction

A player's preferred choice may depend on what others choose.

- Strategic interdependence
- Players, strategies, and payoffs
- Dominant strategies
- Prisoner's Dilemma

Week 2: Sequential Moves

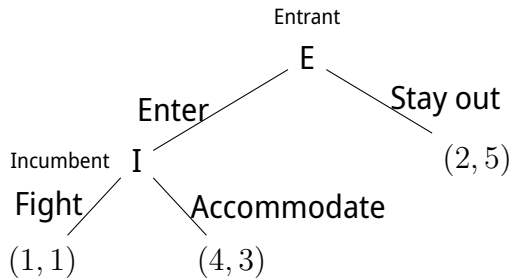
A later player observes an earlier action before responding.

- Decision order and observation
- Game trees
- Backward induction
- Credible responses

The key question is what each player knows when choosing.

How to Read a Game Tree

- A **decision node** shows who chooses.
- A **branch** shows an available action.
- A **terminal node** shows the final payoffs.
- A **path** records one possible sequence of actions.
- Read the decisions forward, but solve them backward.



Observation Check

At each decision node, ask what the player has observed before choosing.

Backward Induction

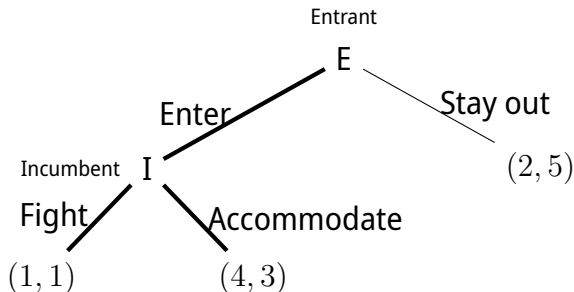
Definition

Begin with the final decisions, determine what later players will choose, and then work backward to the initial move.

- 1 Identify the last player required to make a decision.
- 2 Determine that player's best action at each final decision point.
- 3 Work backward and use those responses to solve the earlier decisions.

Anticipate the response before choosing the move.

Solving the Entry Game



- 1 After entry, the incumbent prefers Accommodate because $3 > 1$.
- 2 The entrant therefore expects entry to produce a payoff of 4.
- 3 Because $4 > 2$, the entrant chooses Enter.

Predicted path: Enter, then Accommodate.

Would the Incumbent Really Fight?

Credible Threat

Carrying out the threatened action would be rational when the decision point is reached.

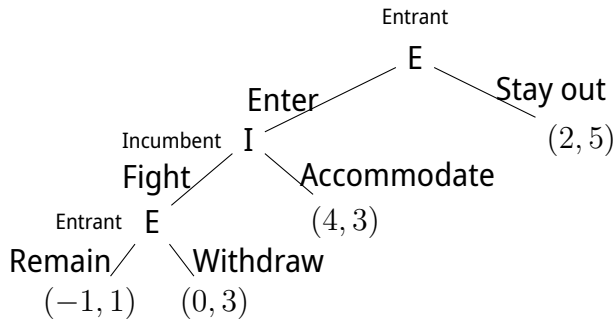
Non-Credible or Empty Threat

The player would prefer not to carry out the announced action when the time came.

Fight gives the incumbent 1; Accommodate gives 3.
Therefore, the threat to fight is not credible.

Commitment may increase credibility by changing later payoffs or making retreat difficult.

Action Versus Strategy: An Extended Entry Game



Action

One choice at one decision node.

Examples: Enter; Withdraw after Fight.

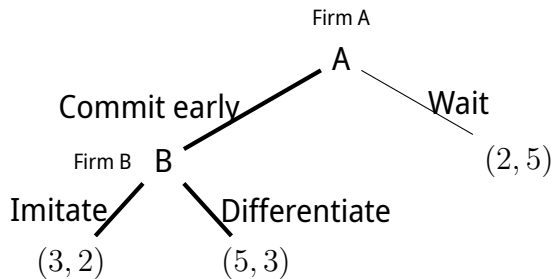
Strategy

A complete plan for every decision point the player might face.

Example: Enter initially; Withdraw if Fight occurs.

An action answers one decision; a strategy answers every

First-Mover Advantage: Shape the Follower's Response



- 1 After an early commitment, Firm B prefers Differentiate: $3 > 2$.
- 2 Firm A then receives 5 from committing early.
- 3 Because $5 > 2$, Firm A commits early.

What Creates the Advantage?

The early commitment establishes a position that induces Firm B to differentiate.

First-Mover Advantage

Moving first helps because the early commitment changes the follower's best response and protects a valuable position.

Second-Mover Advantage: Learn Before Committing

What Waiting Can Provide

- Information about customer demand
- Evidence about technical reliability
- Learning from the pioneer's successes and failures
- Flexibility to improve the product or change direction

Revised Payoff Comparison

- Commit early: payoff 5
- Wait and learn: payoff 6

Because $6 > 5$, Firm A waits.

Second-Mover Advantage

Moving later helps when observation, learning, and flexibility are valuable enough to outweigh the benefits of early commitment.

A Game Tree Is a Simplified Model

A Useful Tree Should Make Clear

- who moves and in what order;
- what each player observes;
- which actions are realistically available;
- how each final outcome is ranked.

Modelling Risk

An incorrect sequence, missing response, or unsupported payoff ranking can produce the wrong strategic prediction.

Simplify the situation, but preserve the incentive that drives the decision.

Application 1: Technology Launch Timing

Discussion Scenario

A firm is deciding whether to launch a new technology early or wait for more information. A rival will observe the decision and choose how to respond.

Discuss the business logic:

- What can an early launch secure?
- What can waiting reveal?
- How might the rival respond?

Build the model:

- Identify the decision order.
- Select two feasible actions for each player.
- Rank the terminal outcomes.
- Draw and solve a simple tree.

Does your tree produce a first- or second-mover advantage, and why?

Application 2: Platform Entry and Regulation

Discussion Scenario

A platform chooses an entry method. The regulator observes that choice and then decides whether to approve, restrict, or challenge it.

Possible entry methods:

- Organic expansion
- Acquisition
- Partnership

Possible regulatory responses:

- Approve
- Approve with restrictions
- Challenge or block

Group Task

Select two entry methods and two regulatory responses. Draw a simple tree, explain the payoff rankings, and solve it by backward induction.

Optional Extension: The Foldable-Phone Decision

Use If Time Allows, or Continue Next Class

A smartphone producer decides whether to launch a foldable phone soon or wait. If it launches soon, an established producer observes the launch and chooses whether to imitate or differentiate.

- 1 Identify the order of moves and what each player observes.
- 2 Draw a simple game tree with feasible actions.
- 3 Explain and rank the terminal outcomes.
- 4 Solve the later player's response first.
- 5 Work backward to recommend Launch Soon or Wait, and identify the source of any timing advantage.
- 6 State one assumption that could reverse your recommendation.

Checklist for the Two Application Trees

Core Criteria

- Correct decision order and observations
- Feasible actions at each node
- Defensible payoff rankings
- Correct backward-induction steps
- Recommendation follows from the tree

Excellent Answers Also

- distinguish evidence from assumptions;
- identify the timing mechanism;
- explain what could reverse the advice.

Final Check

Can another reader follow the decision order, reproduce the backward-induction steps, and identify the assumption driving the recommendation?

Application Debrief

- 1 In each application, what did the later player choose, and why?
- 2 How did that anticipated response change the earlier decision?
- 3 Which payoff assumption created the first- or second-mover advantage?

Managerial Lesson

Timing matters because it changes information, commitment, or later incentives.

Week 2: Main Takeaways

- 1 In a sequential game, later players observe earlier actions before responding.
- 2 A game tree represents players, decision order, actions, paths, and terminal payoffs.
- 3 A strategy is a complete plan, while an action is one choice at one decision point.
- 4 Backward induction solves the final decisions first and then works backward.
- 5 A threat matters only if carrying it out would be rational when the time comes.
- 6 First-mover advantage comes from commitment or position; second-mover advantage comes from information, learning, or flexibility.

Read forward. Solve backward. Explain the timing mechanism.

Exit Ticket and Looking Ahead

Exit Ticket

In two or three sentences, explain why the threat “If you enter, I will fight” may not deter entry. What change in incentives could make the threat credible?

Before Week 3

Complete the assigned pre-reading on simultaneous games and best responses.
Come ready to ask:

How can best responses help us identify stable outcomes when no dominant strategy exists?

Supplementary: Formal Definition of a Subgame

For a Perfect-Information Tree

A subgame begins at a decision node and includes that node, every successor node, every branch following it, and all associated terminal payoffs.

- The entire game is itself a subgame.
- A proper subgame begins after the initial node.
- In games with information sets, a subgame cannot cut through an information set.

The incumbent's decision after entry begins a proper subgame.

Supplementary: Predicted Path Versus Strategy Profile

Predicted Path

The sequence of actions expected to occur:

Enter, then Accommodate.

Strategy Profile

One complete strategy for each player:
(Enter; Accommodate after entry).

The path describes what happens; the strategy profile describes the complete plans.

Supplementary: Subgame-Perfect Equilibrium

Definition

A strategy profile is a **subgame-perfect equilibrium** if it forms a Nash equilibrium in every subgame.

- Planned actions must remain optimal whenever their decision points are reached.
- The requirement applies even after histories that are not expected on the predicted path.
- In finite perfect-information games, backward induction finds a subgame-perfect equilibrium.
- The refinement eliminates equilibria supported by non-credible threats.

Every promised response must pass the later incentive test.

Supplementary: Sources of Mover Advantage

Moving First Can Help When...

early commitment changes the follower's best response;
scarce customers, locations, resources, or partners can be secured;
network effects or standards reward an installed base;
switching costs protect early adoption;

Moving Later Can Help When...

observing demand or technology provides valuable information;
imitation and improvement are relatively inexpensive;
the first mover bears market-education and experimentation costs;
flexibility is valuable and early commitment may become obsolete.

Diagnostic Question

What does moving early secure, and what does waiting allow the firm to learn?

Supplementary: When Should the Tree Include More Players?

Add Another Player Only When That Player Makes a Strategic Choice

A stakeholder belongs in the tree when its action, observation, and incentives materially change the decision paths.

Keep Two Players When...

- customers or suppliers mainly affect payoffs;
- one response drives the focal decision;
- additional branches would not change the recommendation.

Add a Third Player When...

- that player chooses among meaningful actions;
- the action depends on observed earlier moves;
- the response changes later incentives or feasible paths.

Supplementary: Common Errors in Game Trees

Common Error	Correction
Begin with the first mover's preferred action.	Solve the final decision first and carry its consequence backward.
Compare the two payoffs at one terminal node.	At a decision node, compare the moving player's payoff across branches.
Treat an announced threat as credible.	Check whether carrying it out is optimal after the relevant history.
Describe only what happens on the predicted path.	For formal analysis, also specify responses after other possible histories.
Assume moving first is always better.	Identify the commitment, information, scarcity, learning, or flexibility mechanism.